Two positive equilibria outside two disks for a monostable reaction

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Abstract

We construct a positive monostable reaction and two disjoint disks for which an exterior reaction–diffusion equation with zero Dirichlet boundary data has two distinct bounded positive equilibria. This gives a negative answer to the universal two-obstacle uniqueness question posed by Berestycki and Graham. One equilibrium lies above a positive profile on the infinite strip between the disks. A second is obtained by energy minimization on bounded exhaustions; positive transverse energy bounds its mass in a long narrow part of the gap. The two mass bounds are incompatible. The reaction and finite disk geometry are explicit, and both equilibria are classical on the full exterior domain.

1 Introduction

Consider bounded positive equilibria of a reaction–diffusion equation outside absorbing obstacles:

$$-\Delta u = f(u) \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{1}$$

Here the unknown u is a population density and Ω is its habitat. We assume a positive monostable reaction, in the following precise sense:

$$\begin{gathered} f \in C^{1,\gamma}([0,\infty)), \quad 0 < \gamma < 1, \qquad f(0) = f(1) = 0, \\ f'(0) > 0, \quad f'(1) < 0, \qquad f > 0 \text{ on } (0,1), \quad f < 0 \text{ on } (1,\infty). \end{gathered} \tag{2}$$

There is no assumption that the per-capita growth rate $f(s)/s$ decreases. Thus the reaction class is broader than the strong-KPP class studied in [2].

Geometry has a substantial effect on uniqueness in this class. Berestycki and Graham prove uniqueness outside a compact strongly star-shaped obstacle, in particular a single ball [1, Theorem 1.1]. They also explain that two sufficiently separated balls have a unique positive bounded equilibrium [1, Section 8.5]. Their Open Question 7(ii) asks whether uniqueness holds for the complement of two balls without restrictions on their radii or separation. Two nearby balls have no common star center, so the dilation comparison for a single obstacle does not directly apply. Nonuniqueness for other domain geometries, including domains with attached pockets and infinite strips, is already known [1, Sections 5–6]. The issue here is the complement of exactly two balls.

We give a negative answer to the universal uniqueness question.

Main result. *There is one reaction satisfying* (2) *for every* $0 < \gamma < 1$*, and two equal disjoint disks in* $\mathbb{R}^2$*, for which* (1) *has at least two distinct bounded positive solutions in* $C^2(\Omega) \cap C(\overline{\Omega})$*.*

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The reaction, disks and solutions' separating bounds are specified in Theorem 3.1. One admissible example in dimension two suffices to refute the assertion quantified over all dimensions, reactions and disjoint ball pairs. Both solutions are defined on the full exterior domain and vanish on both complete boundary circles.

The construction uses two features of a pair of large disks with a fixed small gap. An infinite straight strip fits between them. A positive one-dimensional profile on this strip, extended by zero, is a subsolution and produces one equilibrium. At the same time, a long central part of the actual gap has uniformly bounded width. We choose the reaction so that the transverse energy is positive for every nonzero profile in this region. For an equilibrium obtained by energy minimization on expanding bounded domains, suppressing the population in the middle of the gap costs energy only in two fixed-length end collars. Its total squared mass in the gap is therefore bounded independently of the gap's length. The equilibrium above the strip profile violates this bound when that length is chosen large.

Section 2 constructs the reaction and proves the transverse estimate. Section 3 fixes the disks and the strip profile, and Section 4 constructs and separates the equilibria. Appendix A gives the regularity, physical boundary and exhaustion details. All functions are real valued, integrals are Lebesgue integrals, and H_0^1 denotes the usual zero-trace Sobolev space.

2 The reaction and a one-dimensional estimate

We first choose a reaction for which a narrow interval has positive energy for every nonzero profile, although a positive profile can still be a subsolution. This distinction will produce the two equilibria.

Put

$$\theta = \frac{1}{4}, \qquad h = \frac{13}{50}, \qquad \varepsilon = \frac{1}{100}, \qquad H(t) = 3t^2 - 2t^3 \quad (0 \le t \le 1),$$

and define, for $0 \le s \le 1$,

$$g(s) = \begin{cases} 0, & 0 \le s \le \theta, \\ H(100s - 25), & \theta < s < h, \\ 1, & h \le s \le 3/4, \\ H(4(1-s)), & 3/4 < s \le 1. \end{cases}$$

Our reaction is

$$f(s) = \begin{cases} g(s) + \varepsilon s(1-s), & 0 \le s \le 1, \\ -\varepsilon(1 - e^{-(s-1)}), & s > 1. \end{cases} \tag{3}$$

The values and first derivatives match at every joining point, since $H(0) = 0$, $H(1) = 1$, and $H'(0) = H'(1) = 0$. Moreover, f and f' are bounded and f' is globally Lipschitz. Consequently $f \in C^{1,\gamma}([0,\infty))$ for every $0 < \gamma < 1$. It satisfies all the conditions in (2), with $f'(0) = \varepsilon$ and $f'(1) = -\varepsilon$.

Let $F(s) = \int_0^s f(t)\,dt$. The estimates needed below are

$$0 \le f(s) \le 2, \qquad F(s) \ge 0, \qquad F(s) \le (s-\theta)_+ + \frac{\varepsilon}{2}s^2 \quad (0 \le s \le 1). \tag{4}$$

Indeed, $0 \le g(s) \le \mathbf{1}_{\{s>\theta\}}$ and $s(1-s) \le s$ on this interval.

Lemma 2.1. *Let $\theta, \mu, \ell > 0$ satisfy $\mu\ell^2 \le 16\theta/3$. Then every nonnegative $v \in H_0^1(-\ell, \ell)$ satisfies*

$$\frac{1}{2}\int_{-\ell}^{\ell} |v'|^2 - \mu \int_{-\ell}^{\ell} (v-\theta)_+ \ge 0. \tag{5}$$

Proof. Use the absolutely continuous representative of v. If $E = \{v > \theta\}$ is empty, there is nothing to prove. Otherwise write $m = |E|$, so $0 < m < 2\ell$. On a component interval I of E, the function $z = v - \theta$ belongs to $H_0^1(I)$. For $I = (a, b)$, set $p(x) = \mu(x - a)(b - x)/2$, so $-p'' = \mu$ and its endpoint values are zero. Completing the square gives

$$\int_I \left(\frac{1}{2}|z'|^2 - \mu z\right) = \frac{1}{2}\int_I |z' - p'|^2 - \frac{1}{2}\int_I |p'|^2 \geq -\frac{\mu^2|I|^3}{24}.$$

Summing over the component intervals, and using $\sum_I |I|^3 \leq m^3$, bounds their total contribution below by $-\mu^2 m^3/24$.

The two outer intervals, from $-\ell$ to $\inf E$ and from $\sup E$ to ℓ, have positive lengths L_- and L_+. Their endpoint values are respectively $0, \theta$ and $\theta, 0$. By Cauchy–Schwarz their gradient energy is at least

$$\frac{\theta^2}{2}\left(\frac{1}{L_-} + \frac{1}{L_+}\right) \geq \frac{2\theta^2}{L_- + L_+} \geq \frac{2\theta^2}{2\ell - m}.$$

All remaining gradient terms are nonnegative. Thus the left-hand side of (5) is at least

$$\frac{2\theta^2}{2\ell - m} - \frac{\mu^2 m^3}{24}.$$

This is nonnegative: the maximum of $m^3(L - m)$ on $[0, L]$ is $27L^4/256$, attained at $m = 3L/4$, and the hypothesis with $L = 2\ell$ is exactly the required inequality. The argument also covers countably many components, since the integrals of both $|v'|^2$ and $(v - \theta)_+$ are finite. □

The following consequence will control the population in the gap. Its width bound is fixed now so that the same estimate applies to every horizontal section of the region constructed in Section 3.

Corollary 2.2. *For the reaction* (3)*, let* $\ell_1 = 9/8$*. If* $0 < \ell \leq \ell_1$ *and* $v \in H_0^1(-\ell, \ell)$ *takes values in* $[0, 1]$*, then*

$$\int_{-\ell}^{\ell} \left(\frac{1}{2}|v'|^2 - F(v)\right) \geq \frac{1}{40}\int_{-\ell}^{\ell} v^2. \tag{6}$$

Proof. Apply Lemma 2.1 with $\mu = 26/25$ and $\theta = 1/4$. Its hypothesis holds because $\mu\ell_1^2 = 1053/800 < 4/3$. Together with (4), this gives

$$\int \left(\frac{1}{2}|v'|^2 - F(v)\right) \geq \frac{1}{52}\int |v'|^2 - \frac{1}{200}\int v^2.$$

The interval Poincaré constant is $\pi^2/(4\ell^2)$. For example, for compactly supported smooth v, write $v = \varphi z$ with $\varphi(x) = \cos(\pi x/(2\ell))$, and integrate by parts to obtain

$$\int |v'|^2 - \frac{\pi^2}{4\ell^2}\int v^2 = \int \varphi^2 |z'|^2 \geq 0.$$

Density gives the same inequality on H_0^1. Since $\pi^2/(4\ell^2) \geq 16/9$, we conclude that

$$\int \left(\frac{1}{2}|v'|^2 - F(v)\right) \geq \left(\frac{4}{117} - \frac{1}{200}\right)\int v^2 = \frac{683}{23400}\int v^2 \geq \frac{1}{40}\int v^2.$$

□

3 Two disks and a strip subsolution

Take

$$\ell_0 = \frac{11}{10}, \qquad T = 20000, \qquad R = 40(T+2)^2 = 16\,003\,200\,160. \tag{7}$$

Let

$$c_\pm = (\pm(R+\ell_0), 0), \qquad \Omega = \mathbb{R}^2 \setminus \big(\overline{B_R(c_+)} \cup \overline{B_R(c_-)}\big). \tag{8}$$

The disks are disjoint and the distance between them is $2\ell_0 = 11/5$. The large radius makes a long part of their gap narrow enough for Corollary 2.2. More precisely, set

$$k(y) = \ell_0 + R - \sqrt{R^2 - y^2} \quad (|y| < R), \qquad D = \{(x,y) : |y| < T,\ |x| < k(y)\}.$$

The lateral sides of D are portions of the two physical circles. For $|y| \le T+2$,

$$\ell_0 \le k(y) = \ell_0 + \frac{y^2}{R + \sqrt{R^2 - y^2}} \le \ell_0 + \frac{y^2}{R} \le \ell_1. \tag{9}$$

Theorem 3.1. *For the reaction f in* (3) *and the two-disk exterior domain Ω in* (8)*, problem* (1) *has two distinct solutions $u, U \in C^2(\Omega) \cap C(\overline{\Omega})$. Both satisfy $0 < u, U < 1$ throughout Ω, and*

$$\int_D u^2\,dx\,dy \le 4500, \qquad \int_D U^2\,dx\,dy > 5000. \tag{10}$$

All parameters in this theorem are finite and fixed. In particular, the theorem is an example in dimension two with strictly separated obstacles; it does not use a limiting domain of parallel walls. The particular numerical values are chosen for convenient estimates, without optimizing the radii or gap.

To construct U, we use the entire strip $\{|x| < \ell_0\}$, which lies inside Ω. Recall that $h = 13/50$ and $f(s) \ge 1$ for $h \le s \le 3/4$. Put

$$q = \frac{11 + \sqrt{17}}{20}, \qquad A = h + \frac{q^2}{2} = \frac{173 + 11\sqrt{17}}{400},$$

and define

$$w(x) = \begin{cases} A - x^2/2, & |x| \le q, \\ q(\ell_0 - |x|), & q < |x| < \ell_0, \\ 0, & |x| \ge \ell_0. \end{cases} \tag{11}$$

Since $q(\ell_0 - q) = h$, both w and w' agree at $x = \pm q$. On $|x| < q$, we have $h \le w \le A < 3/4$, so $f(w) \ge 1 = -w''$. On the linear pieces $-w'' = 0 \le f(w)$. At each of $x = \pm\ell_0$, the derivative jumps upward by q; therefore $-w''$ has a negative Dirac mass there. It follows that

$$W(x,y) := w(x), \qquad -\Delta W \le f(W) \tag{12}$$

in the sense of distributions on the whole plane, and hence on Ω. On both physical circles $W = 0$, because their points have $|x| \ge \ell_0$.

We will also use the strict estimate

$$\int_{-\ell_0}^{\ell_0} w(x)^2\,dx > \frac{1}{8}. \tag{13}$$

Indeed, $q > 3/4$ and $A > 217/400$, so $w(x) \ge A - 1/32 > 1/2$ when $|x| \le 1/4$.

4 Construction of the two equilibria

We work first on bounded domains

$$\Omega_n = B_n(0) \cap \Omega, \qquad n \in \mathbb{N}, \quad n > 2R + \ell_0 + 2,$$

with boundary values zero on the two disks and one on the outer circle. These data have the H^1 extension $b(z) = \min\{1, \operatorname{dist}(z, \Omega^c)\}$. Only the outer radius n will tend to infinity. The reaction, the obstacles and the region D stay fixed.

We use the following bounded-domain Sobolev facts: bounded H^1 sequences on a bounded Lipschitz domain admit weak H^1 and strong L^2 subsequences; traces are continuous; and zero-trace functions are approximable in H^1 by smooth compactly supported functions. These facts, the truncation rules and Poincaré's inequality are recalled in [3, Section 1.1]. The interior regularity and the preservation of the physical boundary values in our exhaustion are detailed in Appendix A.

4.1 An equilibrium above the strip profile

The functions W and 1 are an ordered subsolution and supersolution on every Ω_n, compatible with the prescribed boundary values. Here is the bounded-domain construction. Choose $K \geq \max\{1, \operatorname{Lip}_{[0,1]} f\}$, set $v_0 = W$, and solve

$$(-\Delta + K)v_{j+1} = f(v_j) + Kv_j$$

with the fixed boundary values. The linear Dirichlet problem is solved by minimizing

$$\frac{1}{2}\int_{\Omega_n} (|\nabla v|^2 + Kv^2) - \int_{\Omega_n} (f(v_j) + Kv_j)v$$

in $b + H_0^1(\Omega_n)$. This quadratic is coercive and strictly convex. Its comparison principle follows by testing a weak supersolution against its negative part. Since $s \mapsto f(s) + Ks$ is nondecreasing on $[0, 1]$, comparison gives

$$W \leq v_j \leq v_{j+1} \leq 1.$$

The right-hand sides are uniformly bounded. Testing the linear equations against $v_j - b$ and using Poincaré's inequality bounds v_j in $H^1(\Omega_n)$. Weak compactness, strong L^2 convergence and bounded convergence then give a weak solution U_n with $W \leq U_n \leq 1$. It is classical in the interior.

Lemma A.1 yields a subsequential exterior limit U with the complete zero boundary values and $U \geq W$. As $W \not\equiv 0$, Lemma A.3 gives $0 < U < 1$.

4.2 A minimizing exhaustion

For this construction only, extend f oddly to the negative half-line and let F denote its primitive on $\mathbb{R}$. The extension is C^1, bounded, and has bounded derivative. Its primitive is even, globally Lipschitz, and maximized at ± 1. On Ω_n minimize

$$J_n(v) = \int_{\Omega_n} \left(\frac{1}{2}|\nabla v|^2 - F(v)\right) dx\, dy \quad \text{over } b + H_0^1(\Omega_n).$$

The direct method applies. Since F is bounded above, a minimizing sequence has bounded gradients; Poincaré's inequality then bounds it in H^1. For a strongly L^2-convergent subsequence, the potential integrals converge by

$$\left|\int_{\Omega_n} F(v_j) - F(v)\right| \leq \operatorname{Lip}(F)|\Omega_n|^{1/2}\|v_j - v\|_{L^2(\Omega_n)}.$$

The gradient energy is weakly lower semicontinuous and the trace is preserved, so a global minimizer exists.

Replacing it by $\min\{1, |v|\}$ preserves the boundary data and cannot increase the energy: the gradient decreases and the value of F increases. We may therefore choose a minimizer u_n with $0 \le u_n \le 1$. It is still a minimizer in the *unrestricted* affine class. Thus

$$-\Delta u_n = f(u_n)$$

weakly, and its second variation gives

$$\int_{\Omega_n} \left(|\nabla \phi|^2 - f'(u_n)\phi^2\right) dx\, dy \ge 0 \qquad (\phi \in C_c^\infty(\Omega_n)). \tag{14}$$

The differentiations are justified by the boundedness of f and f'. Interior regularity makes u_n classical. We next obtain a bound in the long gap that is uniform in n.

4.3 An energy bound in the gap

Let $\eta(y)$ be zero for $|y| \le T-1$, one for $|y| \ge T$, and linear in each of the two intervening unit intervals. Then $|\eta'| \le 1$. Inside D, replace u_n by ηu_n, leaving it unchanged elsewhere. This is a legal variation: $(\eta - 1)u_n$ has zero trace on the lateral physical boundary, and on the horizontal cuts $y = \pm T$ because $\eta = 1$. Its zero extension belongs to $H_0^1(\Omega_n)$. Minimality and (4) imply

$$J_D(u_n) \le J_D(\eta u_n) \le \int_C (|\nabla u_n|^2 + 1)\, dx\, dy, \qquad C = \{(x,y) \in D : T-1 < |y| < T\}. \tag{15}$$

Here J_D denotes the same energy restricted to D. For the last inequality we used $F(\eta u_n) \ge 0$ and $\frac{1}{2}|\eta \nabla u_n + u_n \eta' e_y|^2 \le |\nabla u_n|^2 + 1$.

To bound the gradient in these collars, take a Lipschitz function $\chi(y)$ equal to one for $T-1 \le |y| \le T$, zero outside $T-2 < |y| < T+1$, and linear on the remaining two ramps at each end. It satisfies $|\chi'| \le 1$. Use $\chi^2 u_n$ as a test on the central channel $\{|y| < T+2,\ |x| < k(y)\}$, and extend it by zero. It has zero trace on the physical sides and near the horizontal ends, so it belongs to $H_0^1(\Omega_n)$. The weak equation and Young's inequality give

$$\begin{aligned} \int \chi^2 |\nabla u_n|^2 &\le 2 \int \chi^2 f(u_n) u_n + 4 \int u_n^2 |\chi'|^2 \\ &\le 8 \cdot \frac{27}{2} = 108. \end{aligned} \tag{16}$$

In fact, the support has total y-length 6 and width at most $2\ell_1 = 9/4$, hence area at most $27/2$; also $f(u_n)u_n \le 2$. The area of C is at most $9/2$. From (15),

$$J_D(u_n) \le 108 + \frac{9}{2} = \frac{225}{2}. \tag{17}$$

For almost every $y \in (-T, T)$, the horizontal section of u_n belongs to $H_0^1(-k(y), k(y))$: its x-derivative is square integrable by Fubini, and its endpoint values on the physical circles are zero by Lemma A.1. Applying Corollary 2.2 on each such section and retaining the nonnegative y-gradient contribution, we obtain

$$J_D(u_n) \ge \frac{1}{40} \int_D u_n^2\, dx\, dy.$$

Consequently

$$\int_D u_n^2\, dx\, dy \le 4500. \tag{18}$$

4.4 Proof of Theorem 3.1

By Lemma A.1, a subsequence of u_n converges locally in $C^2(\Omega)$ to a solution u with the complete zero boundary values. The stability inequalities (14) and Lemma A.2 exclude the zero limit. Lemma A.3 therefore gives $0 < u < 1$. Since D has finite area and $0 \le u_n \le 1$, dominated convergence in (18) yields

$$\int_D u^2 \, dx \, dy \le 4500.$$

On the other hand, $U \ge W$, and every horizontal section of D contains $(-\ell_0, \ell_0)$. By (13),

$$\int_D U^2 \, dx \, dy \ge 2T \int_{-\ell_0}^{\ell_0} w(x)^2 \, dx > \frac{T}{4} = 5000.$$

Thus $u \ne U$, and both are solutions of the same exterior problem in the required class. This proves the theorem.

A Regularity, physical boundary values and exterior limits

We give the analytic details used in the bounded-domain constructions. They also make explicit why the limiting functions solve the problem on the whole exterior, including both complete physical boundaries.

Lemma A.1. *For the fixed reaction* (3) *and fixed domain* (8)*, let $v_n \in H^1(\Omega_n)$ solve $-\Delta v_n = f(v_n)$, take values in $[0,1]$, and have zero trace on the two inner circles. There is a subsequence converging in $C^2_{\mathrm{loc}}(\Omega)$ to a solution $v \in C^2(\Omega) \cap C(\overline{\Omega})$, with $0 \le v \le 1$ and zero values on both full circles.*

Proof. We first justify the interior estimates for initially weak solutions. On a fixed ball $B_{2r} \Subset \Omega$, write

$$P(z) = \frac{1}{2\pi} \int_{B_{2r}} \log \frac{1}{|z-\zeta|} \, f(v_n(\zeta)) \, d\zeta.$$

The logarithmic kernel and its first derivative are locally integrable in two dimensions. Thus P and ∇P are uniformly bounded on fixed bounded sets, and $-\Delta P = \mathbf{1}_{B_{2r}} f(v_n)$ distributionally. The coefficient $1/(2\pi)$ follows from the flux identity $-\int_{\partial B_\rho} \partial_\nu((2\pi)^{-1}\log(1/|z|)) = 1$. After a compact smooth cutoff, Morrey's inequality at $p = 4$ gives a uniform $C^{0,1/2}$ bound for P. The remainder $v_n - P$ is weakly harmonic and bounded, hence smooth with uniform interior derivative bounds. Here we use [3, Theorem 1.5 and Corollary 1.12].

It follows that v_n, and then $f(v_n)$, have uniform local $C^{0,1/2}$ bounds. Mollify on a smaller ball by a nonnegative smooth kernel of integral one. The smooth mollifications satisfy the Poisson equation with uniformly $C^{0,1/2}$ right-hand sides. The interior Schauder estimate

$$\|a\|_{C^{2,\alpha}(B_{1/2})} \le C_\alpha\big(\|a\|_{L^\infty(B_1)} + \|\Delta a\|_{C^{0,\alpha}(B_1)}\big), \qquad 0 < \alpha < 1,$$

for smooth a, is [3, Theorem 2.12]. Translate and rescale it to the fixed balls, with $\alpha = 1/2$. Letting the mollification scale tend to zero proves uniform local $C^{2,1/2}$ bounds for v_n. Compactness on fixed balls and a countable diagonal selection now give a locally C^2-convergent subsequence, whose limit solves the same equation.

For the boundary, choose $\delta = 1/2$. The annulus $R < |z - c_\pm| < R + \delta$ lies in Ω, since the gap between the two disks is $11/5 > \delta$. It lies in every Ω_n considered here. On this annulus set

$$B(r) = -\frac{r^2 - R^2}{2} + C_R \log(r/R), \qquad C_R = \frac{1 + ((R+\delta)^2 - R^2)/2}{\log((R+\delta)/R)}.$$

Then $-\Delta B = 2$, $B(R) = 0$ and $B(R+\delta) = 1$. The weak maximum principle gives $B \geq 0$. Since $-\Delta(B - v_n) = 2 - f(v_n) \geq 0$ and the boundary traces of $B - v_n$ are nonnegative, the same principle gives $0 \leq v_n \leq B$. This is precisely the weak comparison in [3, Proposition 1.13]. The bound is uniform in n, and $B(r) \to 0$ as $r \downarrow R$. It therefore gives the continuous zero value of the limit at every point of both circles. □

Lemma A.2. *If, in addition, the functions v_n in Lemma A.1 satisfy*

$$\int_{\Omega_n} (|\nabla\phi|^2 - f'(v_n)\phi^2) \geq 0 \qquad (\phi \in C_c^\infty(\Omega_n)),$$

then their exterior limit is not identically zero.

Proof. For each fixed compactly supported test ϕ, pass to the limit in this inequality. If $v \equiv 0$, then $f'(0) = \varepsilon > 0$ would give

$$\int_\Omega |\nabla\phi|^2 \geq \varepsilon \int_\Omega \phi^2 \qquad (\phi \in C_c^\infty(\Omega)).$$

Fix a nonzero smooth function ψ supported in the unit ball. For each $S > 0$, the ball centered at $(0, 4R + 4S)$ with radius S lies in the fixed exterior domain. The translated dilation of ψ into this ball has Rayleigh quotient

$$S^{-2}\frac{\int |\nabla\psi|^2}{\int \psi^2}.$$

Taking S large contradicts the preceding inequality, since $\varepsilon > 0$. □

Lemma A.3. *Every nonzero solution v furnished by Lemma A.1 satisfies $0 < v < 1$ in Ω.*

Proof. Since $-\Delta v = f(v) \geq 0$, the strong minimum principle gives $v > 0$. In this setting the principle follows directly from spherical means: for a nonnegative superharmonic C^2 function, the spherical average about a point with value zero is at most zero; hence the function vanishes on a ball, and then on its connected domain.

For the upper bound put $z = 1 - v \geq 0$, and take $L \geq \max\{1, \mathrm{Lip}_{[0,1]} f\}$. Then $\Delta z = f(v) \leq Lz$. On $\Omega \times (-\pi/(2\sqrt{L}), \pi/(2\sqrt{L}))$, the function $z(x,y)\cos(\sqrt{L}\,t)$ is nonnegative and superharmonic. An interior zero would make it identically zero by the same minimum principle, so $v \equiv 1$. This contradicts its continuous zero values on the physical boundary. □

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References

[1] H. Berestycki and C. Graham, *A Stable-Compact Method for Qualitative Properties of Semilinear Elliptic Equations*, Archive for Rational Mechanics and Analysis **250** (2026), article 21. `https://doi.org/10.1007/s00205-026-02168-6`.

[2] H. Berestycki and C. Graham, *The steady states of strong-KPP reactions in general domains*, arXiv:2212.06611 (2022). `https://arxiv.org/abs/2212.06611`.

[3] X. Fernández-Real and X. Ros-Oton, *Regularity Theory for Elliptic PDE*, author-hosted version, preface dated Zürich 2020. `https://user.math.uzh.ch/ros-oton/Llibre-ellipticPDE.pdf`. The theorem and page locators used here refer to this version. Published by EMS Press, 2022, DOI `https://doi.org/10.4171/ZLAM/28`.